

BusGuard

Autonomous passenger assistance

Singapore BusTech Grand Challenge 2026

IHL Student Category

Team and institution: TBC



3D software preview

Assistance at the bus stop

Passengers need a clear way to ask for help and know what happens next.

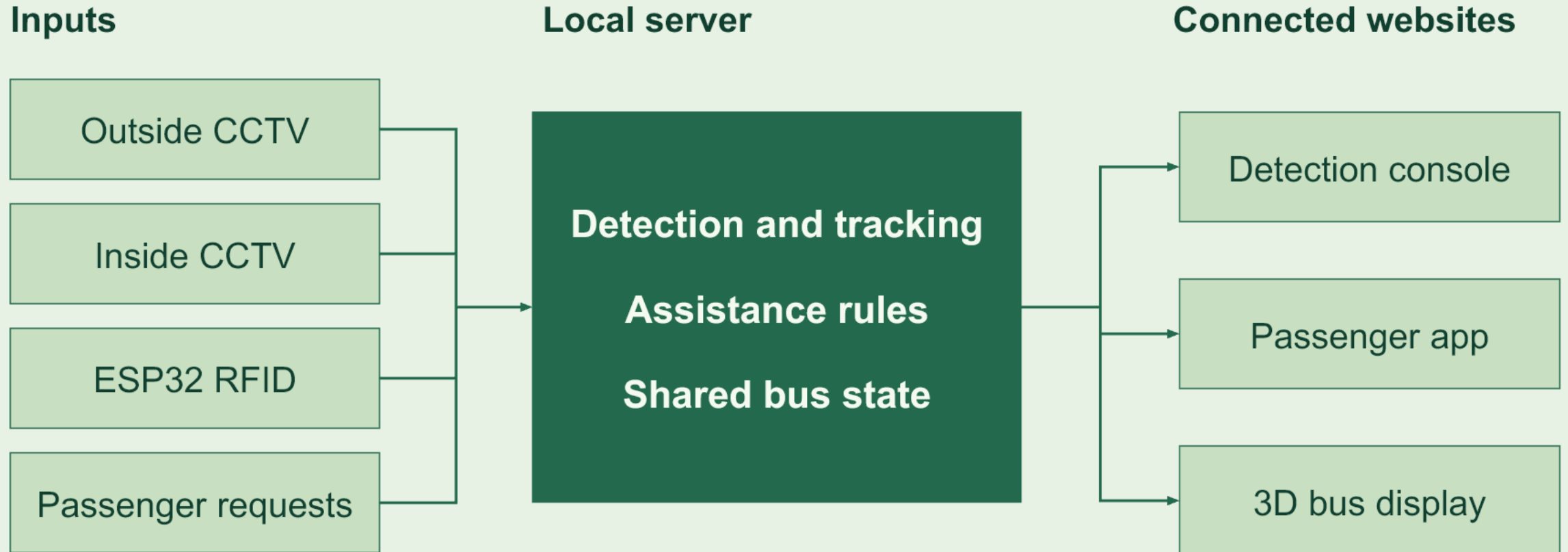
More time to board or alight

Step-free access

Spoken and visible instructions

Passengers declare age-related or pregnancy-related needs through a request or card profile.

One shared assistance controller



Cameras and RFID supply physical inputs. Bus movement, ramp and doors remain simulated.

Boarding time follows activity

15 seconds	Initial quiet window after doors fully open.
10 seconds per RFID tap	Accepted taps protect time from that tap.
20 seconds for ramp access	An accepted passenger request protects its full allowance.
50-second admission cutoff	New requests and taps close. Earlier accepted allowances finish.

Example: a tap accepted at 49 seconds can finish at 59 seconds.

Standing checks and passenger counts

5 seconds

Before departure

Doors closed. Fresh valid checks must report no standing person. Standing or missing evidence restarts confirmation.

30 seconds

Cabin count cross-check

RFID updates the count immediately. The rounded mean of 30 one-second camera samples corrects it. Seat distribution remains estimated.

Inside counting stays active. Outside detection runs only with doors fully open.

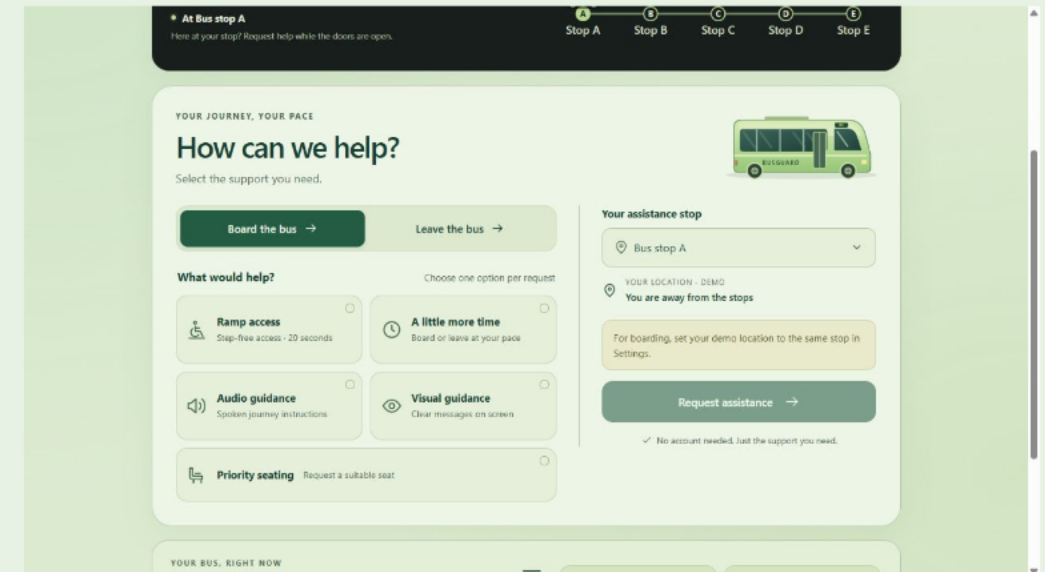
A passenger chooses one request

Ramp access, more time, priority seating, spoken guidance or visual guidance.

A shared countdown explains the accepted allowance.

“Hey BusGuard” activates brief voice help. The reply finishes before listening resumes.

Arrival alerts offer visible messages, optional speech and supported vibration.



Passenger app software preview

Engineering contribution

One authoritative state

The phone, console and bus display share the same decision.

Controlled event handling

Fresh camera evidence, repeat-safe RFID taps and bounded request allowances.

Visible reasons for holds

Standing, missing input and access conditions remain distinct.

YOLOE and OpenAI supply model capabilities. The project integrates the assistance workflow.

Evidence and next validation

Implemented software

Shared controller and three websites
Timing, ownership and retry checks
Camera and RFID integration paths

Next evaluation

Standing errors on labelled cabin scenes
Physical reader and Wi-Fi recovery
Usability with accessibility users

The demonstration uses simulated doors, ramp and motion.
We have not established real-bus accuracy or passenger satisfaction.